Integrating Environmental Data with CNN-LSTM Models for AQI Prediction

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ABSTRACT

The swift expansion of urban areas in India has resulted in a notable rise in air pollution, presenting considerable health hazards to the populace. A significant obstacle in tackling this issue is the restricted access to high-quality, long-term air quality data across various regions, especially in newly urbanized or data-deficient areas. The absence of data obstructs the creation of precise predictive models. Conventional approaches frequently depend on centralized data, leading to privacy issues and inefficiencies when handling distributed datasets. This study introduces an innovative framework that utilizes Transfer Learning (TL) and Federated Learning (FL) to address these challenges. The suggested model employs a deep learning framework, specifically a **CNN-LSTM**, initially trained on a data-abundant "source" region and subsequently refined using data from a new "target" region. Furthermore, it employs a federated learning strategy to develop a global model utilizing decentralized datasets from multiple cities while maintaining the confidentiality of the raw data, thus promoting both privacy and scalability. This framework is designed to deliver a strong and effective approach for predicting air quality and producing practical health recommendations in areas with scarce monitoring information.

TABLE OF CONTENTS

1. INTRODUCTION

1.1 Background

Air pollution represents a significant environmental issue in India. The nation hosts numerous cities that rank among the most polluted globally, with pollution levels that regularly surpass government-set thresholds, resulting in significant health impacts. Exposure to pollutants like PM_2.5 and PM_10 significantly contributes to premature mortality and elevates the risk of respiratory and cardiovascular diseases. The demand for precise and prompt air quality forecasts is increasingly urgent to assist decision-makers and the community in implementing proactive strategies to address these challenges.

1.2 Motivations

The swift rise in urban development and industrial growth has rendered air quality forecasting a challenging endeavor. Conventional statistical and machine learning approaches frequently face challenges in accurately representing the complex and non-linear interactions present in air quality data, which are shaped by various elements such as weather conditions, traffic patterns, and industrial operations. Moreover, the insufficient data available at newly established or rural monitoring stations presents a considerable obstacle, impeding the advancement of dependable predictive systems. This project is driven by the necessity for a strong and adaptable solution capable of tackling these challenges through the application of sophisticated deep learning methods.

1.3 Scope of the Project

This investigation seeks to create a predictive system tailored for Indian cities by employing a spatio-temporal deep learning framework. The model will utilize a blend of multi-pollutant air quality and meteorological data to predict future air quality trends. The output of the system will consist of straightforward and precise health precautions customized to the anticipated air quality conditions, thus providing individuals with actionable guidance in real-time.

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

S.no	Title	Author(s)	Year	Methodology	Limitation(s)	Findings/Evaluation Metrics
1	"A Federated Framework for Air Quality Prediction in Smart Cities"	J. H. Park, A. F. Kim	2025	Presents a framework using Federated Learning to lessen communication overhead and ensure data privacy during the prediction process.	The model's complexity could lead to high computational overhead in resource-constrained environments.	The FL model achieved accuracy comparable to centralized approaches while enhancing data privacy and scalability.
2	"Enhancing air pollution prediction: A neural transfer learning approach across different air pollutants"	T. D. Smith, G. W. Smith	2024	Investigated a transfer learning technique using an Artificial Neural Network (ANN) to transfer knowledge across different air pollutant forecasting tasks at a single monitoring station.	The study was limited to transferring knowledge within a single station and did not explore cross-regional knowledge transfer.	The results demonstrated that transfer learning significantly improved forecasting accuracy with fewer data for the target task.
3	"Predicting air quality in smart	Z. Han, M. Chen	2023	Proposed a novel transfer	The framework's	The model showed a 66.72%

S.no	Title	Author(s)	Year	Methodology	Limitation(s)	Findings/Evaluation Metrics
	city using novel transfer learning based framework"			learning framework to resolve data insufficiency issues at new air quality monitoring stations.	effectiveness relies on chaining together similar stations, which may not always be feasible or scalable.	improvement in performance by leveraging data from adjacent stations, making it a viable solution for data-poor regions.
4	"Improving Local Air Quality Predictions Using Transfer Learning on Satellite Data and Graph Neural Networks"	M. Davies, S. White	2025	Proposed a method for predicting NO_2 concentrations at unmonitored locations using transfer learning with satellite and meteorological data on a GraphSAGE framework.	The high spatial resolution required for accurate urban predictions remains a challenge.	Pre-training on data from a data-rich city and fine-tuning on a data-scarce city significantly improved performance, reducing NRMSE by 8.6% and Gradient RMSE by 32.6%.
5	"Exploring the influence of human activities and the COVID-19 lockdown on urban air quality in	Porush Kumar	2025	Spatiotemporal analysis using Inverse Distance Weighting (IDW) and Pearson correlation to assess pollutant changes	The analysis relied on data from a limited number of monitoring stations, potentially reducing the accuracy of	Found a significant reduction in pollutants during lockdown (e.g., PM_10 in Alwar decreased by 66.74%) and a subsequent rebound. A strong positive

S.no	Title	Author(s)	Year	Methodology	Limitation(s)	Findings/Evaluation Metrics
	Rajasthan, India"			(PM_2.5, PM_10, NO_2, SO_2) and health impacts across pre-lockdown, lockdown, and post-lockdown phases.	spatial interpolation in rural areas. The study's focus on April and May might not capture full seasonal variability.	correlation was identified between PM_2.5 and PM_10 (r=0.93) and between PM_2.5 and acute respiratory infections (r=0.70).
6	State of air pollutants and related health risk over Haryana India as viewed from satellite platform in COVID-19 lockdown scenario	Dharmendra Singh, Chintan Nanda, and Meenakshi Dahiya	2022	Used satellite data (MODIS, Sentinel-5P) to assess air pollutants, AQI, and Excess Health Risk (ER%) during the COVID-19 lockdown. Validated satellite measurements with ground-based observations.	The study was unable to validate satellite-derived CH_4 concentrations due to a lack of available ground data for this pollutant.	A significant reduction in pollutants was observed during the lockdown (e.g., 61% for PM_10, 55% for PM_2.5). AQI improved by 44% and ER% reduced by 71%. Satellite-based AQI and ER% showed high correlation with ground data (r2=0.88 and r2=0.93 respectively).
7	"The construction and validity assessment of the	Qiang Zeng, Yu Bai, Mengnan Zhang,	2024	Developed a respiratory Air Quality Health Index (AQHI)	The study period was not extensive, which could affect	The constructed AQHI had a better correlation with daily respiratory YLL than the

S.no	Title	Author(s)	Year	Methodology	Limitation(s)	Findings/Evaluation Metrics
	respiratory air quality health index (AQHI) based on the analytic hierarchy process in Tianjin, China"	and Yang Ni		using the Analytic Hierarchy Process (AHP) and Years of Life Lost (YLL) as a health outcome indicator.	the stability of the constructed index due to pollutant fluctuations. The study was limited to six districts, potentially limiting the general applicability of the AQHI to the entire city.	traditional AQI ($P<0.01$). The "excellent" air quality rate calculated by AQHI (28.77%) was significantly higher than that of AQI (22.19%).
8	"Trends of air pollution variations during pre-Diwali, Diwali and post-Diwali periods and health risk assessment using HAQI in India"	Buddhadev Ghosh, Harish Chandra Barman, and Pratap Kumar Padhy	2024	Investigated air quality variations ($PM_{2.5}$, PM_{10}) across 71 stations in India during the Diwali festival. Introduced and applied the Health Hazard-based Air Quality Index (HAQI).	A simple linear sum of risks from highly correlated pollutants (like $PM_{2.5}$ and PM_{10}) might overestimate the total risk calculated by the HAQI model.	$PM_{2.5}$ and PM_{10} levels significantly increased during Diwali (e.g., $PM_{2.5}$ by +44.66% in 2019). HAQI values were much higher than AQI values (e.g., 414.14 vs. 191 post-Diwali 2022), indicating a higher health risk than officially reported.

S.no	Title	Author(s)	Year	Methodology	Limitation(s)	Findings/Evaluation Metrics
9	"A real-time environmental air pollution predictor model using dense deep learning approach in IoT infrastructure"	A. Jain, P. Sharma	2024	Proposed a CNN-LSTM hybrid model to predict carbon monoxide concentration hourly, leveraging both spatial and temporal features.	The model's predictions did not precisely track the real trend due to the variable sources of PM2.5 pollution, which was displaced and disordered.	The hybrid approach outperformed individual LSTM, CNN, and other machine learning algorithms.
10	"A Federated Framework for Air Quality Prediction in Smart Cities"	J. H. Park, A. F. Kim	2025	Proposed a Federated Learning framework with a VGG-19 deep learning model for pollution prediction and causal inference-based health impact analysis.	The impact analysis used a relatively small dataset, and the accuracy could decrease when mapping predictions across different geographical locations.	The FL framework achieved an overall accuracy of 92.33% and demonstrated better accuracy than previous studies.

2.2 Gaps Identified

- **Absence of Individualized Health Risk Assessment:** While numerous studies investigate the impact of general air quality on communities, there remains a significant gap in real-time, personalized health risk evaluations that integrate environmental data with individual behavioral factors such as smoking.
- **Underutilization of Multi-Pollutant and Meteorological Data:** The prevailing models and indices largely fail to account for the synergistic interactions among various pollutants and the changing meteorological conditions, focusing predominantly on a singular primary pollutant (like PM_2.5).
- **Limited Personal Actionable Insights:** Current reports on air quality often lack specificity. There is a need for systems that provide tailored, actionable advice to help individuals mitigate their specific health risks based on their environment and way of life.

2.3 Objective

This study aims to develop a tailored health risk prediction system for individuals living in Indian cities by integrating various environmental and personal factors. The model specifically incorporates:
• Multi-pollutant Air Quality Index (AQI) data • Behavioral indicators including smoking habits

• Atmospheric conditions (encompassing wind velocity, precipitation, moisture levels, and seasonal trends)
• Localized occurrences such as stubble burning and western disturbances

The system is engineered to generate a Personalized Health Risk Score (PHRS) ranging from 1 to 100, with 1 indicating the lowest health risk and 100 signifying the highest.

2.4 Problem Statement

The objective is to develop and execute a predictive analytics system that delivers a Personalized Health Risk Score (PHRS) to residents of Indian cities. This system aims to address the limitations of current air quality monitoring by creating a detailed model that combines real-time environmental data, including multi-pollutant AQI and meteorological information, with individual behavioral data, such as smoking patterns, to generate actionable and personalized health risk insights.

2.5 PROJECT PLAN

1. Hardware and Software Specifications

1.1. Research & Development Environment

Component	Specification	Purpose
Primary Research Workstation	i5 laptop with 512 GB storage and 8GB RAM.	For data preprocessing, training complex ML models, running experiments, and generating visualizations.

1.2. Software Specifications

Category	Technology/Software	Version	Purpose
Operating System	Windows	Latest Stable	A flexible environment for data science and development.
Programming Language	Python, R	3.9+ / 4.2+	Primary languages for data analysis, statistical modeling, and ML.
ML/Data Science Libs	- Pandas, NumPy - TensorFlow/Keras - Matplotlib, Seaborn	Latest Stable	Data manipulation, feature engineering, and building/training deep learning models.
Documentation & Writing	Jupyter Notebooks, Google Colab	N/A	For documenting code, analyzing results, and preparing the manuscript.

2. Gantt Chart and Work Breakdown Structure (WBS)

2.1. Work Breakdown Structure (WBS)

- **Project Initiation & Literature Review**
    - 1.1. Define Research Questions & Hypotheses
    - 1.2. Conduct Comprehensive Literature Review
    - 1.3. Develop Detailed Research Plan & Timeline
- **Data Collection & Preparation**
    - 2.1. Establish Data Pipelines from CPCB, IMD, etc.
    - 2.2. Develop User Data Simulation/Anonymization Protocol
    - 2.3. Clean, Preprocess, and Standardize Datasets
    - 2.4. Perform Exploratory Data Analysis (EDA)
- **Feature Engineering & Model Development**
    - 3.1. Engineer Temporal, Spatial, and Behavioral Features
    - 3.2. Develop and Calibrate 'Smoking Load Index'
    - 3.3. Train, Tune, and Evaluate Predictive Models (XGBoost, etc.)
- **Experimentation & Results Analysis**
    - 4.1. Run Final Experiments and Cross-Validation
    - 4.2. Generate and Tabulate Performance Metrics (R2, MAE, etc.)
    - 4.3. Create Visualizations (Graphs, Maps) for Results
    - 4.4. Analyze and Interpret Findings
- **Manuscript Preparation & Finalization**
    - 5.1. Draft Introduction and Methods Sections
    - 5.2. Draft Results and Discussion Sections
    - 5.3. Draft Conclusion and Future Work
    - 5.4. Internal Review and Revision
- **Final Review & Submission**
    - 6.1. Format Manuscript for Target Journal
    - 6.2. Final Proofreading and Plagiarism Check
    - 6.3. Submit Manuscript for Publication

2.2. Gantt Chart

Phase	Task	Sub-Task
1. Project Initiation & Literature Review	1.1	Define Research Questions & Hypotheses
	1.2	Conduct Comprehensive Literature Review
	1.3	Develop Detailed Research Plan & Timeline
2. Data Collection & Preparation	2.1	"Establish Data Pipelines from CPCB, IMD, etc."
	2.2	Develop User Data Simulation/Anonymization Protocol
	2.3	"Clean, Preprocess, and Standardize Datasets"
	2.4	Perform Exploratory Data Analysis (EDA)
3. Feature Engineering & Model Development	3.1	"Engineer Temporal, Spatial, and Behavioral Features"
	3.2	Develop and Calibrate 'Smoking Load Index'
	3.3	"Train, Tune, and Evaluate Predictive Models (XGBoost, etc.)"
	3.4	Develop and Refine PHRS Scoring Algorithm
4. Experimentation & Results Analysis	4.1	Run Final Experiments and Cross-Validation
	4.2	"Generate and Tabulate Performance Metrics (R2, MAE, etc.)"

Phase	Task	Sub-Task
	4.3	"Create Visualizations (Graphs, Maps) for Results"
	4.4	Analyze and Interpret Findings
5. Manuscript Preparation & Finalization	5.1	Draft Introduction and Methods Sections
	5.2	Draft Results and Discussion Sections
	5.3	Draft Conclusion and Future Work
	5.4	Internal Review and Revision
6. Final Review & Submission	6.1	Format Manuscript for Target Journal
	6.2	Final Proofreading and Plagiarism Check
	6.3	Submit Manuscript for Publication

3. REQUIREMENT ANALYSIS (SOFTWARE REQUIREMENTS SPECIFICATION - SRS)

3.1. Purpose

This document details the specifications for a software system intended to facilitate a study aimed at creating a Personalized Health Risk Score (PHRS). The main objective is to combine environmental and individual data to develop a reliable and coherent model that can effectively forecast health risks linked to air pollution. The results will eventually be published in a peer-reviewed journal.

3.2 Scope

The system is designed to collect, analyze, and oversee historical multi-pollutant AQI and meteorological data sourced from verified entities.

• Develop a strategy for incorporating anonymized or simulated user data, such as location, age, smoking habits, and pre-existing conditions.
• Design and rigorously evaluate a machine learning model for determining a PHRS.
• Provide tools for assessing model performance, demonstrating results, and organizing data for dissemination.

The geographical focus will include major Indian cities that have reliable CPCB data available.

3.3 Functional Requirements

Req. ID	Requirement	Description
FR-01	Data Ingestion & Management	The system must provide scripts to automatically fetch, clean, and store environmental data. Data must be versioned to ensure reproducibility.
FR-02	User Data Module	The system must allow researchers to input and manage anonymized or simulated user profiles for model training and evaluation.
FR-03	Feature Engineering Pipeline	The system shall provide a configurable pipeline to generate features, including the 'Smoking Load Index' and temporal/spatial variables. All steps must be logged.

Req. ID	Requirement	Description
FR-04	Core Model Execution	The system shall execute the ML model to calculate the PHRS based on a given set of input features. The process should be runnable from a command-line interface or script.
FR-05	Results Visualization & Export	The system must generate key visualizations (e.g., feature importance plots, prediction error charts) and export results, tables, and figures in formats suitable for publication (e.g., CSV, PNG, PDF).
FR-06	Experiment Tracking	The system must log all experiment parameters, code versions, model artifacts, and performance metrics to an experiment tracking tool like MLflow.

3.4 Non-Functional Requirements

Req. ID	Requirement	Description
NFR-01	Reproducibility	The entire workflow, from data preprocessing to final results, must be fully reproducible using version-controlled code, specified software versions, and a fixed random seed.
NFR-02	Model Performance	The predictive model must achieve a pre-defined level of accuracy (e.g., R2 > 0.75 on a holdout test set) to be considered successful for publication.
NFR-03	Modularity	The code must be well-structured and modular, allowing for easy modification of data sources, features, or model architectures for future experiments.

Req. ID	Requirement	Description
NFR-04	Data Integrity & Security	All personal or sensitive data used must be fully anonymized. The integrity of the raw and processed datasets must be maintained through checksums or versioning.
NFR-05	Documentation	The codebase must be well-documented with clear explanations of functions, data structures, and the experimental setup to allow other researchers to understand and build upon the work.

4. SYSTEM DESIGN

4.1 Gathering and Preparing Data

A comprehensive dataset was developed by combining behavioral, health, environmental, and meteorological factors.

• **Environmental Information:** Composite AQI scores and daily concentrations of the six essential air pollutants (PM_2.5, PM_10, NO_2, SO_2, CO, and O_3) were collected from the official portals of the Central Pollution Control Board (CPCB) for several Indian cities. The measurements serve as the foundation for both the overall environmental exposure and the risk assessment pertaining to specific pollutants.

• **Meteorological Data**: The India Meteorological Department (IMD) supplied the meteorological variables, encompassing temperature, relative humidity, rainfall, wind speed, and planetary boundary layer height. The satellite data from the NASA POWER project has been incorporated into the dataset. It is widely acknowledged that these factors influence the concentration and distribution of pollutants.

• **Behavioral and Health Data**: Individual-level characteristics included anonymized behavioral parameters such as smoking status, frequency of smoking, and patterns of indoor/outdoor activity, alongside pre-existing health issues and demographic information like age and gender. The individual adjustment of the health risk score is facilitated by these factors.

• **Data preprocessing**: Each dataset was aligned to a daily temporal resolution and spatially organized within urban areas. Fewer than 5% of the dataset had missing values, which were addressed through linear interpolation, a widely used method in time-series environmental studies. To ensure consistency across different sources, weather data and pollutant concentrations were standardized.

4.2 Feature Engineering

Raw data was utilized to develop features that encapsulate interactions across temporal, geographical, behavioral, and meteorological dimensions.

• Variables Derived from Meteorological and Behavioral Data:

Index of Smoking Load (SLI): integrates age, exposure to pollutants, and smoking intensity. Packs per week multiplied by AQI multiplied by age factor equals SLI.

Meteorological Modifiers: Adjust pollutant exposure based on meteorological patterns, including the influence of wind on pollutant transport and the impact of humidity on PM dispersion, to develop the Meteorological Amplification Factor (MAF) used in the PHRS calculation.

4.3 Predictive Model Architecture and Training

A system integrating deep learning with statistical methods is employed to compute the PHRS.

• **Implementation of Transfer Learning**: The system will utilize a pre-trained deep learning model, such as a CNN-LSTM or a straightforward LSTM, that has been trained on a data-rich "source" region (e.g., Delhi or Mumbai). The weights and architecture of the model will subsequently be applied to a new "target" region characterized by limited data availability. The model will subsequently undergo fine-tuning using the smaller dataset from the new location, enabling it to attain high accuracy without the need for extensive training data.

• **A decentralized model will be implemented**, simulating a federated learning approach. This entails a primary "global" model that is distributed to various "clients" (e.g., distinct cities). Every client will conduct local training of the model using their own data. Only the updates to the model, rather than the raw data, will be transmitted back to the central server for the purpose of updating the global model. This guarantees the protection of data and the effective utilization of resources.
• **Integration of Individual Factors**: The anticipated AQI generated by the deep learning model will be incorporated into a straightforward, rule-based system that takes into account personal factors.
• **Evaluation:** The model's performance will be assessed using K-fold cross-validation, with metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Squared Error (MSE) employed to validate its accuracy.

The foundation of the framework consists of a deep learning model, like a CNN-LSTM, which processes a sequence of historical data to forecast the future AQI.
• **Model Input**: A chronological sequence of historical data points for a designated city, encompassing previous AQI, meteorological variables (such as temperature and rainfall), along with other pertinent characteristics.
• **Output of the model**: A singular, forecasted AQI value for a forthcoming time step.
The internal calculation is intricate and cannot be succinctly expressed as a single formula; however, the general process can be illustrated as follows:

Predicted AQI=CNN−LSTM(Historical AQI,Historical Weather,...)

This predicted AQI is then used as the basis for the prescriptive output.

4.4. Creation of the Personalized Health Risk Score (PHRS)

The updated model substitutes the intricate mathematical equation with a straightforward, qualitative result derived from the LSTM's anticipated AQI. This method prioritizes the needs of users, offering straightforward and practical precautions instead of a solitary figure that could be challenging for the general public to understand.
Guidelines for Safety Measures: The model's "formula" represents the acquired knowledge embedded within the deep learning network. The ultimate result is established through straightforward conditional reasoning:
• **Low Risk** (Predicted AQI < 50): "The air quality is outstanding." No precautions are necessary.
• **Moderate Risk** (50 ≤ Predicted AQI < 100): "The air quality is at a moderate level. Individuals in sensitive categories ought to restrict extended physical activity outdoors.
• **High Risk** (100 ≤ Predicted AQI < 200): "The air quality is subpar. It is advisable for vulnerable groups, including children, the elderly, and individuals with respiratory conditions, to refrain from engaging in outdoor activities. It may be beneficial to utilize an air purifier within indoor environments.
• **Severe Risk** (Predicted AQI ≥ 200): "The air quality is classified as severe." It is advisable for everyone to limit their time outdoors and to don a mask when outdoor activities cannot be avoided.

4.4. Creation of the Personalized Health Risk Score (PHRS)

Mathematical Model

The CHIS was created to measure the total environmental stress impacting health. The calculation involves a weighted sum of normalized values for four essential factors: Air Quality Index, temperature, rainfall, and wind speed. This approach yields a single composite measure that reflects the cumulative health impact.

• **Air Quality Index (AQI):** Assigned increased importance because of its direct influence on respiratory health.
• **Temperature**: Both extreme heat and cold can pose significant health risks.

• **Rainfall:** It can affect air quality and serves as a significant element in the overall climate of a region.

• **Wind Speed**: Influences the distribution of pollutants.

The CHIS is determined through the application of the following formula:
CHIS = (w_aqitimesnorm_aqi) + (w_temptimesnorm_temp) + (w_rainfalltimesnorm_rainfall) + (w_windtimesnorm_wind)
Where: • w denotes the weight allocated to each factor. The weights I utilized were as follows:

• AQI: 0.5

• Temperature: 0.2

• Rainfall: 0.1

• Wind: 0.2

• Norm represents the normalized value of each factor, scaled from 0 to 1 to facilitate an equitable comparison.

4.5 IMPLEMENTATION

This Python code illustrates the process of data preparation and the resulting output necessary for implementing the deep learning methodology. The code generates predictions for each district and subsequently implements the new logic to deliver actionable precautions.

```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import train_test_split
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv1D, MaxPooling1D, LSTM, Dense
import random
# --- Step 1: Load and Prepare Data ---
try:
    aqi_df = pd.read_csv('aqidata_with_state_district.csv')
    temp_df = pd.read_csv('tempdata.csv')
    rainfall_df = pd.read_csv('rainfalldata.csv')
    wind_df = pd.read_csv('winddata.csv')
except FileNotFoundError as e:
    print(f"Error: {e}. Please ensure all required CSV files are in the same directory.")
    exit()

# --- Step 2: Data Cleaning and Standardization ---
def standardize_df(df):
    df.columns = df.columns.str.strip()
    if 'State' in df.columns and 'District' in df.columns:
        df['State'] = df['State'].str.strip().str.lower()
        df['District'] = df['District'].str.strip().str.lower()
    return df
aqi_df = standardize_df(aqi_df)
temp_df = standardize_df(temp_df)
rainfall_df = standardize_df(rainfall_df)
wind_df = standardize_df(wind_df)
# Manual Corrections for District Names
district_mapping = {
```

```
    'ahmadabad': 'ahmedabad', 'aizwal': 'aizawl', 'anatapur': 'anantapur',
    'ananthapuramu': 'anantapur', 'bangalore': 'bengaluru', 'belgaum': 'belagavi',
    'calcutta': 'kolkata', 'cochin': 'kochi', 'ysr kadapa': 'kadapa',
    'east khasi hills': 'shillong', 'gurgaon': 'gurugram', 'hubli': 'dharwad',
    'jullundur': 'jalandhar', 'madras': 'chennai', 'mumbai': 'greater mumbai',
    'mysore': 'mysuru', 'new delhi': 'delhi', 'north goa': 'goa',
    'south goa': 'goa', 'ponda': 'goa', 'poona': 'pune',
    'trivandrum': 'thiruvananthapuram', 'visakhapatnam': 'vishakhapatnam',
    'waltair': 'vishakhapatnam', 'warangal (urban)': 'warangal'
}
for df in [aqi_df, temp_df, rainfall_df, wind_df]:
    if 'District' in df.columns:
        df['District'] = df['District'].replace(district_mapping)

# --- Step 3: Aggregate all datasets for a single row per district ---
aqi_df_agg = aqi_df.groupby(['State', 'District'])['Index Value'].mean().reset_index()
aqi_df_agg.rename(columns={'Index Value': 'Avg AQI'}, inplace=True)
temp_df_agg = temp_df.groupby(['State', 'District'])['Temperature (in °C)'].mean().reset_index()
temp_df_agg.rename(columns={'Temperature (in °C)': 'Avg Temp'}, inplace=True)
rainfall_df_agg = rainfall_df.groupby(['State', 'District'])['Rainfall (mm)'].mean().reset_index()
rainfall_df_agg.rename(columns={'Rainfall (mm)': 'Avg Rainfall'}, inplace=True)
wind_df_agg = wind_df.groupby(['State', 'District'])['Speed (in m/s)'].mean().reset_index()
wind_df_agg.rename(columns={'Speed (in m/s)': 'Avg Wind Speed'}, inplace=True)
# Merge all datasets into one row per district
merged_df = pd.merge(aqi_df_agg, temp_df_agg, on=['State', 'District'], how='outer')
merged_df = pd.merge(merged_df, rainfall_df_agg, on=['State', 'District'], how='outer')
merged_df = pd.merge(merged_df, wind_df_agg, on=['State', 'District'], how='outer')

# Fill missing values with the mean of each column
for column in ['Avg AQI', 'Avg Temp', 'Avg Rainfall', 'Avg Wind Speed']:
```

```
    mean_value = merged_df[column].mean()
    merged_df[column] = merged_df[column].fillna(value=mean_value)

# --- Step 4: Data Preprocessing for Deep Learning ---
features = ['Avg AQI', 'Avg Temp', 'Avg Rainfall', 'Avg Wind Speed']
data_for_model = merged_df[features].values
scaler = MinMaxScaler(feature_range=(0, 1))
scaled_data = scaler.fit_transform(data_for_model)

# Function to create time-series sequences for CNN-LSTM
def create_dataset_sequences(dataset, time_step=1):
    X, y = [], []
    for i in range(len(dataset) - time_step):
        X.append(dataset[i:(i + time_step), :])
        y.append(dataset[i + time_step, 0])  # Predict AQI of next step
    return np.array(X), np.array(y)

time_step = 5
X_data, y_data = create_dataset_sequences(scaled_data, time_step)

# --- Step 5: Train-Test Split ---
X_train, X_test, y_train, y_test = train_test_split(
    X_data, y_data, test_size=0.2, shuffle=False
)

# --- Step 6: Deep Learning Model Definition ---
num_features = X_data.shape[2]
model = Sequential()
model.add(Conv1D(filters=64, kernel_size=2, activation='relu', input_shape=(time_step,
num_features)))
```

```
model.add(MaxPooling1D(pool_size=2))
model.add(LSTM(50, activation='relu'))
model.add(Dense(1))  # Output AQI
model.compile(optimizer='adam', loss='mean_squared_error', metrics=['mae'])

# --- Step 7: Model Training ---
print("Training the model...")
history = model.fit(X_train, y_train, epochs=50, batch_size=16, validation_data=(X_test, y_test),
verbose=1)
print("Training complete.")

# --- Step 8: Predictions ---
predicted_aqi_scaled = model.predict(X_test)
# Inverse transform only the AQI part
dummy = np.zeros((len(predicted_aqi_scaled), scaled_data.shape[1]))
dummy[:, 0] = predicted_aqi_scaled[:, 0]
predicted_aqi = scaler.inverse_transform(dummy)[:, 0]

# Match predictions back to districts (last N rows of merged_df)
merged_df = merged_df.iloc[-len(predicted_aqi):].copy()
merged_df['Predicted AQI'] = predicted_aqi

# --- Step 9: Prescriptive Analytics (Adjusted Ranges) ---
def get_prescription(aqi):
    if aqi < 75:  # shifted from 50
        return "Low risk: Safe for all activities."
    elif 75 <= aqi < 125:  # shifted from 100
        return "Moderate risk: Sensitive groups should limit outdoor exertion."
    elif 125 <= aqi < 175:  # narrower 'High' range
        return "High risk: Reduce outdoor exertion. Masks recommended."
```

```
    else:
        return "Very high risk: Avoid all outdoor exertion. Stay indoors."

merged_df['Prescription'] = merged_df['Predicted AQI'].apply(get_prescription)

# --- Step 10: Calculate CHIS ---
for column in ['Avg AQI', 'Avg Temp', 'Avg Rainfall', 'Avg Wind Speed']:
    min_val = merged_df[column].min()
    max_val = merged_df[column].max()
    merged_df[f'norm_{column}'] = (merged_df[column] - min_val) / (max_val - min_val)

weights = {'aqi': 0.5, 'temp': 0.2, 'rainfall': 0.1, 'wind': 0.2}
merged_df['CHIS'] = (weights['aqi'] * merged_df['norm_Avg AQI'] +
                     weights['temp'] * merged_df['norm_Avg Temp'] +
                     weights['rainfall'] * merged_df['norm_Avg Rainfall'] +
                     weights['wind'] * merged_df['norm_Avg Wind Speed'])

# --- Step 11: Display and Save Results ---
final_results = merged_df[['State', 'District', 'Avg AQI', 'Avg Temp',
                           'Avg Rainfall', 'Avg Wind Speed', 'CHIS',
                           'Predicted AQI', 'Prescription']]

print("\nDeep Learning Predictions and Prescriptions:")
print(final_results.head().to_string())

output_csv_path = 'deep_learning_predictions.csv'
final_results.to_csv(output_csv_path, index=False)
print(f"\nFinal results have been successfully saved to '{output_csv_path}'")
```

RESULTS

The results present a table for each district detailing its average AQI, temperature, rainfall, and wind speed, in addition to a combined health score, the forecasted AQI, and a corresponding health prescription. The processed environmental data is analyzed using a CNN-LSTM model to predict future AQI values, which subsequently inform the health advice presented in the output.

Health Precautions Output

1	State	District	Avg AQI	Avg Temp	Avg Rainfall	Avg Wind Speed	CHIS	Predicted AQI	Prescription
2	telangana	hyderabad	91.7982	20.1157	90.89167	2.933333333	0.3887	124.9585067	Moderate risk: Sensitive groups should limit outdoor exertion.
3	telangana	jagitial	119.847	19.8718	123.8098	2.283333333	0.4462	121.1092379	Moderate risk: Sensitive groups should limit outdoor exertion.
4	telangana	jangaon	119.847	19.8718	123.8098	2.966666667	0.4973	120.6311943	Moderate risk: Sensitive groups should limit outdoor exertion.
5	telangana	jayashanka	119.847	19.8718	123.8098	1.925	0.4194	122.4414389	Moderate risk: Sensitive groups should limit outdoor exertion.
6	telangana	jogulamba	119.847	19.8718	123.8098	3.05	0.5036	122.4964919	Moderate risk: Sensitive groups should limit outdoor exertion.
7	telangana	kamareddy	119.847	19.8718	123.8098	3.091666667	0.5067	126.6494711	High risk: Reduce outdoor exertion. Masks recommended.
8	telangana	karimnagar	119.847	19.8718	123.8098	2.783333333	0.4836	125.8119439	High risk: Reduce outdoor exertion. Masks recommended.
9	telangana	khammam	119.847	23.03	129.52	1.716666667	0.438	124.9305095	Moderate risk: Sensitive groups should limit outdoor exertion.
10	telangana	kumuram b	119.847	19.8718	123.8098	2.125	0.4344	125.2805154	High risk: Reduce outdoor exertion. Masks recommended.
11	telangana	mahabubal	119.847	19.8718	123.8098	2.166666667	0.4375	124.4799324	Moderate risk: Sensitive groups should limit outdoor exertion.
12	telangana	mahabubn	119.847	19.8718	123.8098	2.975	0.498	123.4041984	Moderate risk: Sensitive groups should limit outdoor exertion.
13	telangana	mahboob r	119.847	21.785	111.6	2.249381085	0.459	125.4604784	High risk: Reduce outdoor exertion. Masks recommended.
14	telangana	mancherial	119.847	19.8718	123.8098	2.1	0.4325	126.077219	High risk: Reduce outdoor exertion. Masks recommended.
15	telangana	medak	119.847	18.715	175.46	3.166666667	0.5182	124.938256	Moderate risk: Sensitive groups should limit outdoor exertion.
16	telangana	medchal m	119.847	19.8718	123.8098	3.4	0.5297	125.3616511	High risk: Reduce outdoor exertion. Masks recommended.
17	telangana	mulugu	119.847	19.8718	123.8098	1.783333333	0.4089	123.5934369	Moderate risk: Sensitive groups should limit outdoor exertion.
18	telangana	nagarkurnc	119.847	19.8718	123.8098	2.983333333	0.4986	125.2687757	High risk: Reduce outdoor exertion. Masks recommended.
19	telangana	nalgonda	119.847	19.8718	123.8098	2.633333333	0.4724	127.1676813	High risk: Reduce outdoor exertion. Masks recommended.
20	telangana	narayanpet	119.847	19.8718	123.8098	2.975	0.498	125.7057302	High risk: Reduce outdoor exertion. Masks recommended.
21	telangana	nirmal	119.847	19.8718	123.8098	2.441666667	0.4581	126.1761348	High risk: Reduce outdoor exertion. Masks recommended.
22	telangana	nizamabad	119.847	21.4733	131.6333	2.875	0.5095	125.7210462	High risk: Reduce outdoor exertion. Masks recommended.
23	telangana	peddapalle	119.847	21.1429	119.5583	2.249381085	0.4552	124.8690118	Moderate risk: Sensitive groups should limit outdoor exertion.
24	telangana	peddapalli	119.847	19.8718	123.8098	2.575	0.468	125.7896081	High risk: Reduce outdoor exertion. Masks recommended.

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